



RESEARCH ARTICLE

Thermal Front Detection Using Satellite-Derived Sea Surface Temperature in the Northern Indian Ocean: Evaluation of Gradient-Based and Histogram-Based Methods

M. Jishad¹  · Neeraj Agarwal¹

Received: 23 June 2021 / Accepted: 19 February 2022 / Published online: 14 March 2022
© Indian Society of Remote Sensing 2022

Abstract

Two different methods of detecting oceanic thermal fronts using satellite-derived high-resolution sea surface temperature (SST) data are evaluated in this study. High-resolution SST observations from INSAT, MODIS and the group of high-resolution SST (GHRSSST) have been used to identify thermal fronts in the Northern Indian Ocean. The thermal fronts are identified using gradient-based and histogram-based techniques. Several sensitivity studies were conducted to determine various thresholds required to identify thermal fronts from both methods. It is found that the detected fronts using gradient-based method are noisy and more in number as compared to histogram-based edge detection technique. The edge detection method can detect prominent fronts with fewer false alarms. Front detection techniques were also applied on sub-daily SST images obtained from geostationary satellite, INSAT-3D. Winter time fronts were realistically detected by using both algorithms. Buoy observations confirmed the presence of detected fronts in the satellite images. Application of the two techniques of front detection on SST images during cyclone shows that the histogram-based method successfully detects thermal fronts associated with cooling. The gradient-based method missed most of the thermal fronts during the cyclone, mainly due to diffused gradients captured in the satellite based merged SST under cloudy conditions.

Keywords Thermal fronts · Gradient · Cyclone · Sea surface temperature · Remote sensing

Introduction

Thermal fronts are the boundaries separating regions of contrasting water mass. These are associated with temperature gradients caused by various physical phenomena in the oceans such as upwelling, eddy formation and other air-sea interaction mechanisms. The thermal front regions are highly rich in nutrients and act as biological indicators in the ocean. Frontal regions exhibit large variations in physical and chemical properties in ocean water mass. Fronts have different life span ranging from few days and up to year and most of them are seasonal (Belkin & Cornillon, 2003). Apart from being large scale ($\sim > 100$ km), there are fronts having sub-mesoscale

variability with length scales of the order 0.1–10 km (Mahadevan, 2016; Thomas et al., 2013). High-resolution satellite observations of sea surface temperature (SST) have been used to study thermal front characteristics, their seasonal and inter-annual variations in the regions of the coastal and marginal seas in the Pacific Ocean (Belkin & Cornillon, 2003, 2004, 2005). Although fronts are very common in the coastal waters, open ocean fronts are also not uncommon and they are present in many parts of the ocean. These fronts can vary rapidly or remain stable for long periods and over long distances depending upon their formation mechanism. These small-scale fronts are formed under favourable mesoscale buoyancy gradients and currents (McWilliams, 2017) and are associated with sub-mesoscale currents that dissipate energy and result in diapycnal mixing (McWilliams, 2016). Some fronts are associated with strong vertical motions on one or both sides, which may contribute to the vertical flux of nutrients and thus to the higher phytoplankton biomass (Marra et al., 1990).

✉ M. Jishad
jishadm@sac.isro.gov.in

¹ Oceanic Sciences Division/AOSG/EPSC, Space Applications Centre-ISRO, Ahmedabad, India

Thermal front identification forms the basis of any potential fishing zone detection (PFZ) algorithm (Jishad et al., 2019; Suhadha & Asriningrum, 2020; Zainuddin et al., 2004), and hence detecting them with accuracy is of utmost importance. There are several methods of detecting thermal fronts from satellite images, using the gradient, variance, skewness (Cornillon & Watts, 1987), or cluster-shade algorithms (Holyer & Peckinpaugh, 1989) or histogram algorithms (Cayula & Cornillon, 1992, 1995).

In this study, an analysis has been performed to choose suitable thermal front detection method for PFZ identification. To this end, we have tested two different approaches, which are largely used in many studies, for thermal front detection from satellite SST images. One approach is based on thermal gradient as proposed by Kostianoy et al. (2004), and the other one is the histogram-based edge detection method by Cayula and Cornillon (1992). The two methods of front detection have been applied on the satellite SST products from Group for High-Resolution Sea Surface Temperature (GHRSSST), Moderate Resolution Imaging Spectroradiometer (MODIS) and INSAT-3D, and the results are inter-compared in the Northern Indian Ocean (NIO). The motivation for this work was to choose a suitable algorithm for the identification of thermal fronts in the NIO that could be then used in the PFZ algorithm developed by Jishad et al. (2019).

The manuscript is organized as follows: Sect. 2 describes the study area followed by a description of various data used in the study in Sect. 3. Section 4 describes the two methods of front detection used in the study and salient results are presented in Sect. 5. Conclusions are provided in Sect. 6.

Study Area

NIO is taken as the study area to demonstrate the relative effectiveness of the two front detecting algorithms [Longitude: 55 °E–95 °E and Latitude: 17 °N–23 °N]. The northern Bay of Bengal (NBOB) is marked by the presence of very strong thermal fronts during the winter season and many rivers are bringing fresh water (Mathur et al., 2019; Sengupta et al., 2016). The eastern coast of India is the favourable region for coastal upwelling (Johns et al., 1992, 1993) due to which there is formation of fronts. The northern Arabian Sea (NAS) is well-mixed and active during Arabian Sea cyclones (Weller et al., 2002). The average SST in the AS is warm enough to support the development of tropical cyclones throughout the year (Evan & Camargo, 2011). However, the monsoon wind and strong vertical wind shear limits cyclone development and intensification and is restricted only during the pre-monsoon and post-monsoon period. The reduction in the SST is

instantaneous in response to the strong wind-mixing, and the subsequent upward Ekman pumping even after the dissipation of cyclone (Price, 1981; Pudov, 1992; Sadhuram, 2004), which leads to the formation of strong thermal fronts.

Data Used

Three SST products from different sources (INSAT3D, GHRSSST and MODIS) were used to compare both front detection methods for a period of three months (January–March 2020). The following sub-sections describe these datasets. The identified thermal fronts were compared with the in situ observations from moored buoys.

INSAT 3D/3DR

Indian satellite for met-ocean applications, INSAT-3D (in geostationary orbit) has a six-band imager paired with a 19-band sounder. SST-1DVAR product is derived with a spatial resolution of 4 km from split thermal window channels (10.2–11.3 μm and 11.5–12.5 μm) during day-time and using additional mid-IR window channel (3.7–4.1 μm) during night time over cloud-free oceanic regions (Gangwar & Thapliyal, 2020; Walton et al., 1998). The 1DVAR SST has a bias of -0.36 K and standard deviation of 0.63 K.

INSAT-3D provide images at every 30-min interval and it has a definite advantage (Agarwal et al., 2019) over polar-orbiting sensors, which provide observations in a specific region, only two times a day. In this study, thermal fronts are computed using the 30 min snapshots as well as daily averaged image of SST from INSAT-3D. The SST data sets were downloaded from MOSDAC (Meteorological and Oceanographic Satellite Data Archival Centre, <https://mosdac.gov.in/>).

MODIS

MODIS has five bands in infrared, mid-infrared and far-infrared, which were placed to optimize their use for SST determination. MODIS SST is retrieved using the line-by-line numerical radiative transfer code developed at Rutherford Appleton Laboratory in the UK as a basis for modelling atmospheric absorption and emission processes in the MODIS infrared bands: (Llewellyn-Jones et al., 1984; Zavody et al., 1995). The daily averaged MODIS-Aqua SST (DegC) having 4 km spatial resolution is downloaded from https://podaac-tools.jpl.nasa.gov/drive/files/allData/modis/L3/docs/modis_sst.html. The accuracy of MODIS SST products is explained in detail by Kilpatrick and et al. (2015).

GHR SST

The Group for High-Resolution Sea Surface Temperature (GHR SST) data merges SST observations from all kinds of available sources. It is a merged SST product that includes SST from all the available microwave and infra-red sensors. The Level-4 (L4) product is generated using various objective analysis techniques to produce gap-free SST maps over the global oceans. The SST product used in this study is generated by Jet Propulsion Laboratory (JPL) using an optimum interpolation approach on a global 0.011° grid (Reynolds et al., 2007). This is version 4 multiscale ultrahigh resolution (MUR) product that has inputs from all possible sources of SST such as NASA Advanced Microwave Scanning Radiometer-EOS (AMSRE), the Moderate Resolution Imaging Spectroradiometer (MODIS) on the NASA Aqua and Terra platforms, the US Navy microwave WindSat radiometer and in situ SST observations from the NOAA iQuam project. GHR SST Products blended from multiple satellites or multiple passes of satellites. The data downloaded from (<https://podaac-tools.jpl.nasa.gov/drive/files/allData/ghrsst/data/GDS2/L4/GLOB/JPL/MUR/v4.1>).

The thermal fronts are detected using GHR SST – $1 \text{ km} \times 1 \text{ km}$, MODIS – $4 \text{ km} \times 4 \text{ km}$, INSAT3D – $4 \text{ km} \times 4 \text{ km}$. Later, adopted methods were also tested for the remapped $4 \text{ km} \times 4 \text{ km}$ GHR SST for better comparison with MODIS and INSAT 3D.

Moored Buoys

The National Institute of Ocean Technology-Ocean Observation Systems (NIOT-OOS) and Indian National Centre for Ocean Information Services (ESSO-INCOIS) are responsible for operating, maintaining and developing moored buoy observational and related telecommunication networks in Indian waters (Venkatesan et al, 2013, 2016). The Indian moored buoy network is designed for understanding the dynamics of air-sea interactions and ocean circulation in the NIO. The Indian moored buoy system continuously records vertical profiles of temperature, salinity and currents up to 500 m depth, which supplements the Research Moored Array for African-Asian-Australian Monsoon Analysis and Prediction (RAMA). The SST data from deep-water moorings in the NAS (AD06, 18.50°N ; 67.44°E) and NBOB (BD09, 17.466°N ; 89.144°E) were used for comparative studies.

Methodology

Edge Detection Algorithm (Cayula & Cornillon, 1992)

In the edge detection algorithm (Cayula & Cornillon, 1992), [Here after it is used as CC92], the SST data are filtered using a Median filter to remove noise from the images. This filtering is performed by using a moving window (3×3) through all the pixels in the image, with the centre of the window replaced by the median of all the other values in the window. This leads to removal of noise from the original data.

In the next step, the entire filtered data are segmented into many windows and each window is processed independently from others to evaluate the probability of an edge being present in that window. The appropriate pixel window size has been chosen by considering the size of the data set. Unwanted features like clouds and other edges may get included while using large window size, while the temperature will be constant for a smaller window. If the transition between the two fronts occupies too much of the window, the two regions of different temperature cannot be identified, and hence the edge goes undetected. The size of the window is fixed as 32×32 pixels and the histogram from each window of filtered data is used to compute pixel values and their frequencies. There can be windows without any edge (unimodal) and with one edge (bimodal). The Histogram algorithm must determine the number of populations present in each window using different thresholds (τ , should be within the data range). If the number of populations are two, the parameters of the mixture are calculated for each population to check the statistical relevancy of the segmentation of the data into two classes. The single listed class indicates that no edge is present within the window. The best threshold τ_{best} (Cayula & Cornillon, 1992) is computed and used for each window.

Considering the spatial distribution of temperature, the possibility of a front existing in that window is ruled out where the pixels of any value are distributed evenly across the window. The cohesion coefficient is used to measure if two distinct populations are present in the window (32×32). The cohesion coefficient is the ratio of the number of times a pixel of one population is immediately adjacent to another pixel and the total number. ‘Adjacent’ could be on all four sides, however, we count only the bottom and right neighbours to avoid redundancy from next pixels.

The last step in this algorithm is to locate the edge pixels in each window if the presence of a front in the given window has been confirmed. We define an edge pixel as a pixel of one population that is immediately adjacent to a

pixel of another population. The output of this step will be an image where all the non-edge pixels have been assigned a value of zero and the edge pixels carry the value of the threshold temperature.

Gradient Method (Kostianoy et al., 2004)

The fronts were also identified using gradient-based approach (Kostianoy et al., 2004), [Here after it is used as KO04], in the NIO. In this technique, first of all the magnitude of spatial temperature gradients ($\text{Grad}_{x,y}$) is computed using the approach of Kostianoy et al. (2004) as;

$$\text{Grad}_{x,y} = \sqrt{\left(\frac{\partial \text{SST}}{\partial x}\right)^2 + \left(\frac{\partial \text{SST}}{\partial y}\right)^2} \quad (1)$$

These values are computed at every grid point of the data and only values above $0.04^\circ\text{C}/\text{km}$ are considered as significant gradients for the region of interest (Shown in Sect. 5.1). The fronts are the boundaries where the maximum gradient values along a gradient feature exist. So to determine this, a search is carried in horizontal and meridional directions and once a gradient feature is encountered the algorithm searches for the maximum value of the gradient uphill along the feature. Once the maximum value of the gradient is found then that point is marked as a front (Mathur et al., 2007). This is continued at all grid points, and we get the thermal fronts marked.

Results and Discussion

The fronts were identified using both CC92 and KO04 methods for GHRSSST (for both 1 km and 4 km horizontal resolutions), INSAT3D and MODIS data.

Significance of Threshold in the Algorithms

For the CC92 algorithm the fronts were identified using histogram analysis and a cohesive algorithm using suitable threshold for bimodal distributions $\theta(\tau)$. The value of $\theta(\tau)$ is the main criteria that determine the existence of a front. It can also be treated as a way to control the sensitivity of our algorithm. The smaller $\theta(\tau)$ might accidentally mark fronts within a single population; and by choosing a $\theta(\tau)$ greater than this value, we can make the algorithm detect only the stronger fronts. The algorithm is tested using different $\theta(\tau)$ from 0.5 to 0.85, and results are shown in Fig. 1. The number of identified thermal fronts decreases with increase in the value of $\theta(\tau)$ and we can change the value of $\theta(\tau)$ for optimal detection of thermal fronts. The value of $\theta(\tau)$ should be greater than 0.75 for bimodal distributions (Cayula, 1988). The figure shows that the value of

$\theta(\tau)$ can be used as 0.75, which can detect fronts with fewer noise.

In the gradient-based method, the temperature gradients were computed for each grid using Eq. 1. The gradient magnitude varies from 0 to $0.12^\circ\text{C}/\text{km}$ for the BoB. To identify frontal lines, significant gradient were required to be identified. This was done using a gradient threshold, which was selected after applying different thresholds ranging from 0.01 to $0.1^\circ\text{C}/\text{km}$ on the computed SST gradients. Figure 2 shows the frontal lines over SST images after applying various thresholds. The number of detected fronts reduce with increasing threshold. In this case we found that a threshold of $0.04^\circ\text{C}/\text{km}$ was optimum for our region, and henceforth, this threshold has been used to determine significant gradients for the rest of this study.

Inter-Comparison of Detected Thermal Fronts From the Two Methods

The front detection algorithms, CC92 and KO04, were used to detect thermal fronts using GHRSSST (1 km and 4 km), INSAT3D and MODIS data for 1–6th January, 2020. The intercomparison of thermal fronts detected from these two methods was performed, and the results are shown in this section.

For GHRSSST – 1 km, thermal fronts were identified using both KO04 and CC92 algorithms and are shown in Fig. 3a, b. It can be seen that KO04 detects most of the prominent thermal fronts using the threshold $0.04^\circ\text{C}/\text{km}$. On the other hand, the CC92 algorithm detects too many fronts, which may be corrected by increasing the value of $\theta(\tau)$. The isolated frontal points (less than 5 points) from KO04 were also filtered.

Both methods were also applied on the $4\text{ km} \times 4\text{ km}$ GHRSSST, and plots are shown in Fig. 3c, d. The frontal lines plotted using the two methods are consistent with the features seen in the SST image.

For INSAT-3D, the detected thermal fronts are shown in Fig. 3e, f. KO04 method is highly sensitive and a small difference in the temperature across pixels can be detected as a front. Hence there are lot of noisy features seen in the fronts detected using KO04 method (even after filtration). However, CC92 detects only prominent and possible fronts.

Large-scale fronts that are seen in the GHRSSST (especially along the coastal region) data are missing in INSAT-3D. This could be due to less spatial variability of INSAT-3D SST (as seen in Fig. 3) in the region. The cold water plume along the east coast of India is more prominent in the GHRSSST observations as compared to the INSAT-3D data which might have resulted in larger spatial gradients in that region in the former case.

Fig. 1 GHRSSST overlaid with the identified thermal fronts (black lines) using different thresholds of $\theta(\tau)$ from 0.5 to 0.85 for 14/01/2020

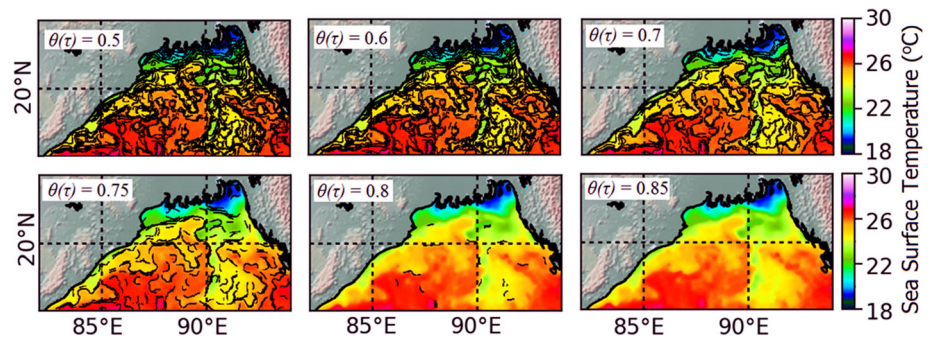


Fig. 2 GHRSSST overlaid with the identified thermal fronts using different gradient thresholds from 0.02 to 0.08 for 14/01/2020

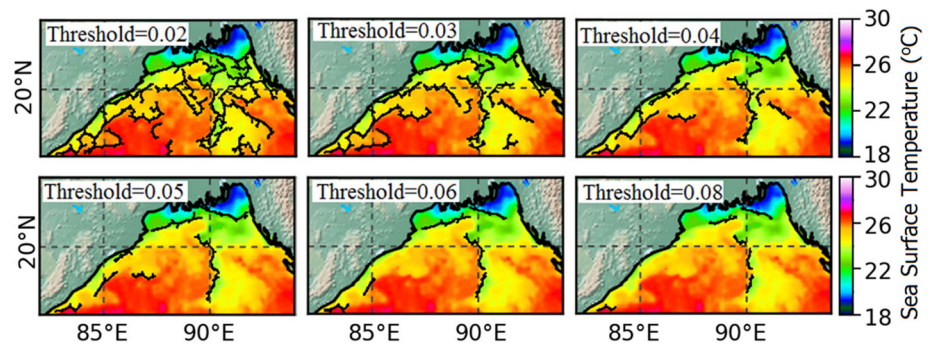
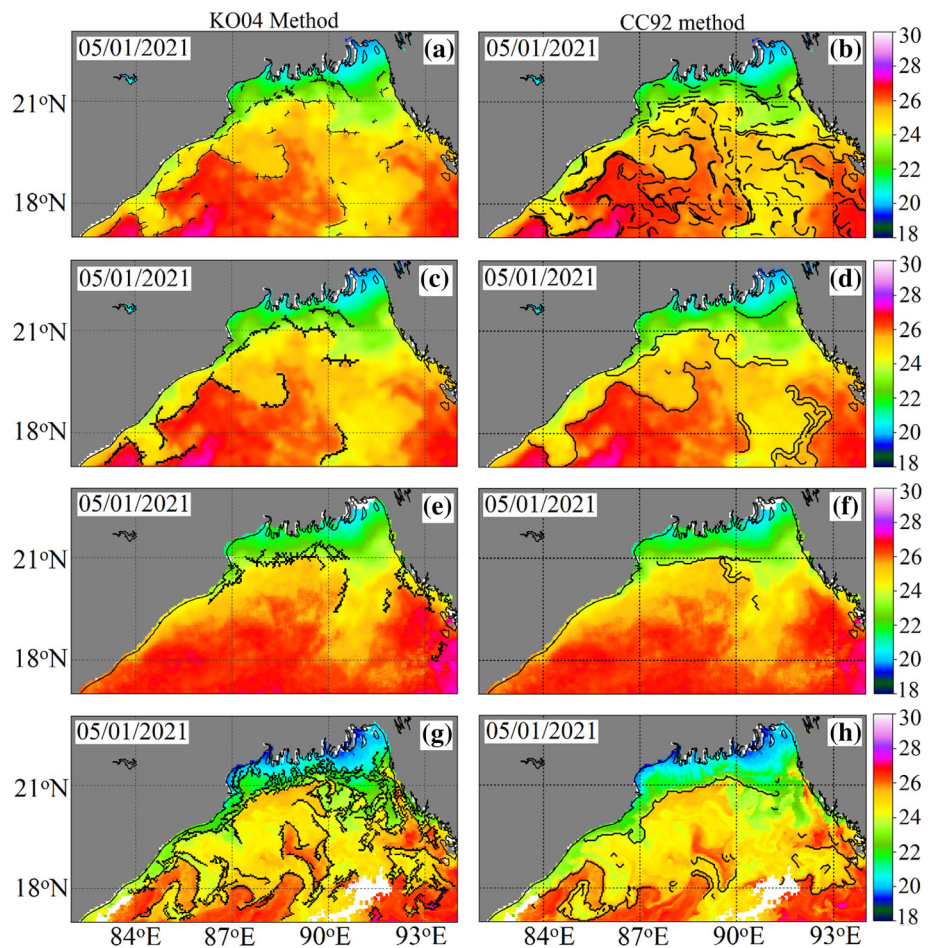


Fig. 3 Thermal fronts obtained by the KO04 method and CC92 method using GHRSSST-1 km (a, b), GHRSSST-4 km (c, d), INSAT-4 km (e, f) and MODIS-4 km (g, h) data



The thermal fronts were also detected using swath data from MODIS and are shown in Fig. 3g, h. This was done to check how the fronts appear in an image from a single satellite pass. One can see that KO04 detected many isolated fronts, which may be regarded as noise, especially in the north east BoB, while the fronts detected using CC92 are quite smooth and realistic.

In the above analysis, daily images were used for the detection of the thermal fronts. In case of INSAT-3D, 48 half hourly images were averaged to generate daily SST. This averaging would have possibly lead to the smoothening of features and hence we saw that number of fronts detected from INSAT-3D daily data were less as compared to the GHRSSST or MODIS data.

Figure 4 shows the application of the two front detection algorithms on the INSAT-3D acquisitions within the same day. Here 4 such snapshots corresponding to 00, 06, 12 and 18 Z are shown for 15 January 2021. For comparison purpose, daily averaged SST for the same day from INSAT-3D is also shown. One can see that both the methods detect almost similar fronts in these images. It can also be seen that there is a slight variation in the detected fronts during the course of the day, which may be attributed to the presence of clouds in some of the acquisitions. The thermal fronts in the daily averaged image seem to

represent the features that are present during different time of the day.

Frontogenesis During Winter Months

The dry cold winds blowing from the north causes large scale significant cooling over the northern Indian Ocean, which, together with freshening caused by river discharge results in the formation of thermal fronts (Sengupta et al., 2016). Figure 5 shows the time sequence of SST images overlaid with thermal fronts (using both CC92 (Fig. 5a) and KO04 (Fig. 5b) algorithms) from 19–22 February 2020. It can be seen that there are several mesoscale fronts (size ~ 100 –250 km) detected by both methods for the given period. Incidentally, one of the fronts, elongated and meridional was in the close vicinity of the Bay of Bengal deep-water moorings (BD08, 17.805°N; 89.24°E and BD09, 17.466°N; 89.144°E). In the satellite observations, this particular front (parallel to 80°E longitude) formed the boundary of water mass having a temperature difference of ~ 1 °C. This front first appeared on 19 February and became elongated on 20 February, by 21st February the front moved slightly eastwards and came quite close to BD08 mooring. By 22 February, the KO04 front was still prevalent, while the frontal line in CC92 vanished.

The time series of SST from BD09 is shown in Fig. 5c. At BD09, the average SST was 26.5 °C on 19 February 2020. The cold water from central BoB was pushed away by the warm water coming from the west coast, which resulted in a warming of 0.7 °C in BD09. Both methods (CC92 and KO04) were able to detect these prominent fronts in the satellite images (Fig. 5a, b), which demonstrates that both the methods seem to be good as far as capturing of mesoscale fronts is concerned.

Frontogeneses During and After Cyclone

The performance of both frontal detection methods was also tested during two different cyclones. The super cyclonic storm Kyarr (24/10/2019 to 02/11/2019) and extremely severe cyclonic storm Maha (30/10/2019 to 07/11/2019) were formed in the NAS. Super Cyclonic Storm Kyarr originated as a low-pressure system near th Lakshadweep Islands on October 17. The system intensified to a tropical storm on October 24 and super cyclonic storm on October 27. The cyclone passed near one of the AS deep moorings (AD06, 18.50°N; 67.44°E). The data from the mooring on 27 October revealed a drop in the atmospheric pressure from 1007 to 996hpa and an increase in the wind speed by 12 m/s. The corresponding reduction in SST of 2.5 °C (Fig. 6) was observed at the buoy location, which was roughly 200 km away from the cyclone eye.

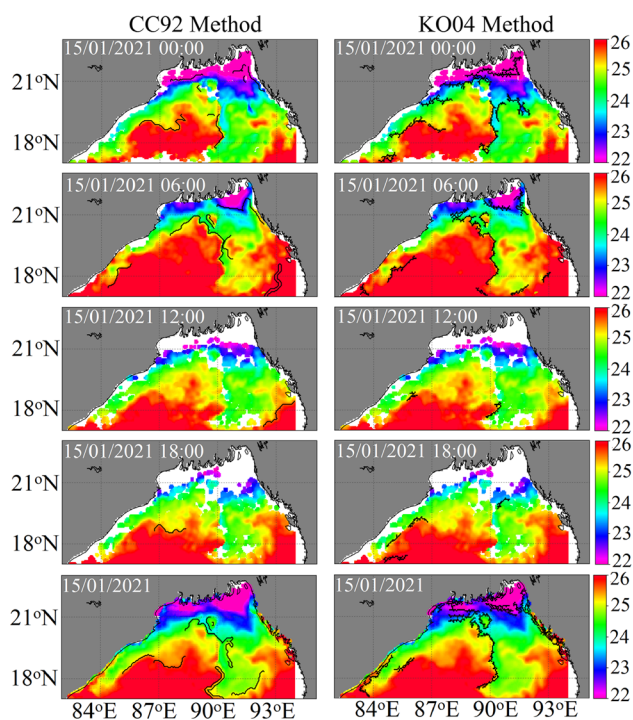


Fig. 4 Thermal fronts detection using CC92 (left panels) and KO04 (right panels) from INSAT-3D SST images at different hours of the day (15/01/2021). Bottom panel shows the thermal fronts from daily averaged INSAT-3D image for 15/01/2021

Fig. 5 Identified thermal fronts from **a** KO04 and **b** CC92 methods for GHRSSST (19–22 February 2020). The white and black dots are the locations of BD08 and BD09 moorings, respectively, **c** Time series of the surface temperature plot from BD09 shows the sharp temperature gradient (solid line) while crossing thermal front

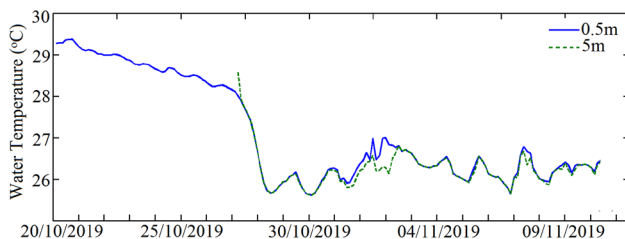
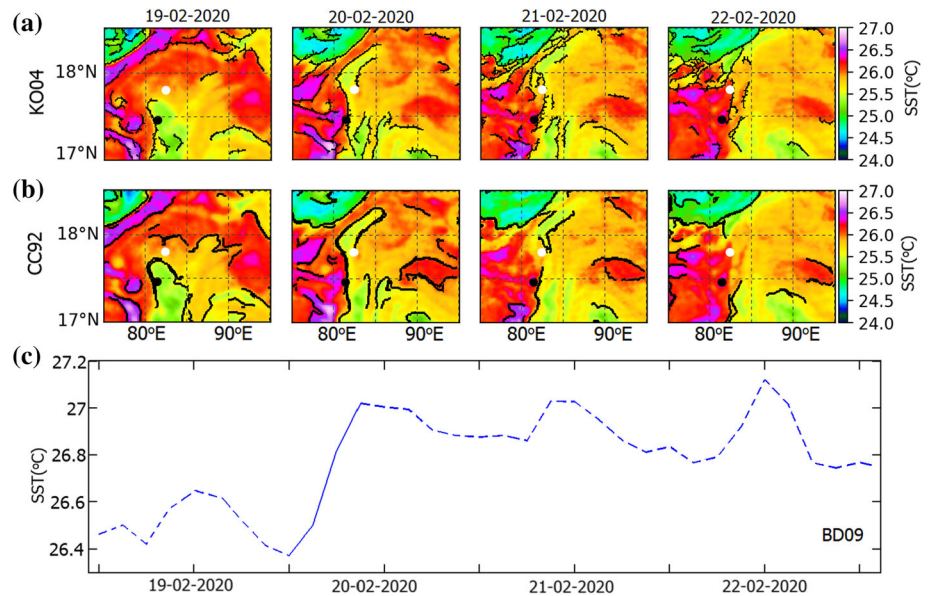


Fig. 6 Time series of water temperature at 0.5 and 5 m water depth recorded at the AD06 mooring in the Arabian Sea

An extremely severe cyclonic storm Maha, developed on 30 October, followed almost a similar westward track like Kyarr until central AS (19.7°N; 63.6°E). After that the track got changed from west to east on 05 November. The cyclone Maha passed by the AD06 buoy twice while moving away and towards Indian mainland. It can be seen that the surface and subsurface temperature (until 20 m) of the ocean decreased rapidly by ~ 2.5 °C (Fig. 6) just after the passage of the cyclone and the cooling persisted many days after the cyclone. GHRSSST observations also captured the intense cooling (Fig. 7), which was mainly caused by the vertical mixing and evaporative cooling due to strong winds. The SST observations showed the elongated patch of cooling all along the path of Kyarr.

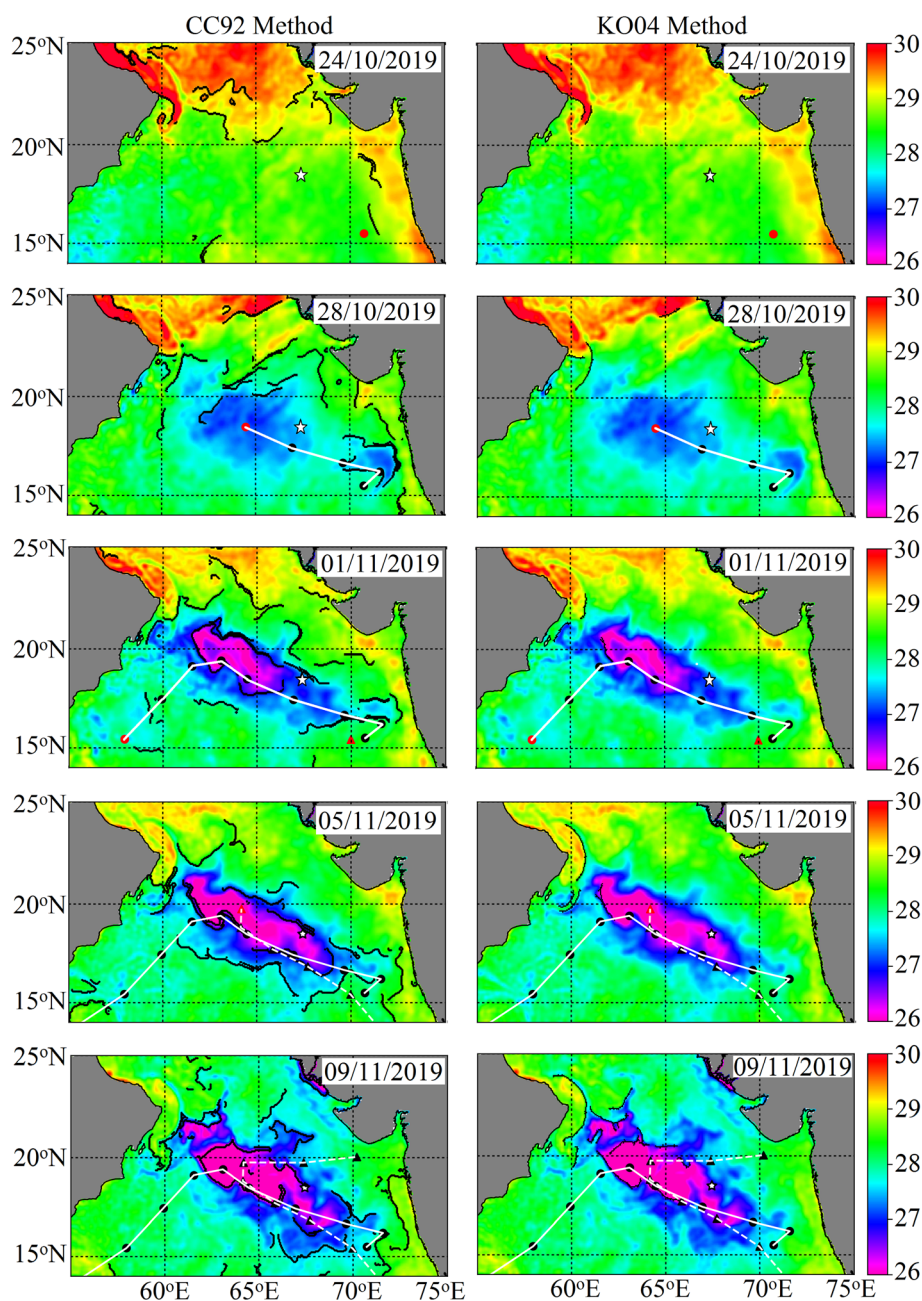
The two techniques of front detection were applied to GHRSSST during this period to evaluate their effectiveness during extreme events. The detected thermal fronts using both KO04 and CC92 methods overlaid on GHRSSST during the period of both the cyclones are shown for different dates in Fig. 7. It can be seen that despite large spatial gradients in SST images due to intense cyclonic cooling KO04 missed most of thermal fronts, which were detected by CC92. One of the reasons of this ineffectiveness of

KO04 method during cyclone time could be the larger contribution of low resolution SST (from microwave sensor) in the GHRSSST product. Thermal gradients in the low-resolution SST are smoothened out and hence no fronts are detected above the desired threshold value of 0.04 °C/km in the KO04 method.

Conclusions

In this study, two different methods, gradient-based (KO04) and histogram-based edge detection method (CC92) were applied on high-resolution satellite SST images obtained from INSAT, GHRSSST and MODIS to detect thermal fronts. Various thresholds to optimally deduct thermal fronts were applied in the two techniques. The thresholds were fixed for bimodal distributions as $\theta(\tau) = 0.75$ and gradient threshold as 0.04 °C/km for CC92 and KO04, respectively. The number of detected fronts is more using KO04, but, CC92 detects only the prominent and possible fronts. Furthermore, the two front detection techniques were applied to SST images during cyclone event. It was found that while the CC92 method was able to detect fronts associated with cyclonic cooling quite remarkably, KO04 failed to identify those fronts. Diffused gradients, due to the blending of low-resolution SST in GHRSSST product seem to be a limitation in the KO04 method of front detection during extreme weather conditions. Temporal frequency of polar orbiting satellites is not sufficient to observe the transient changes in small scale fronts. Geostationary satellites (such as INSAT3D/3R) provide SST observations at 15 min temporal resolution (when used together), which will be very useful to study

Fig. 7 Thermal fronts obtained by the KO04 method and CC92 method using GHRSSST-1 km. The dots connected with white continuous line and triangles connected with white dash line are the track of Kyarr and Maha cyclones, respectively. The red colour dot and the triangle show the position of the cyclone for the corresponding date and the white star is pointing to the location of AD06 buoy (Color figure online)



the fast evolving and decaying thermal fronts. This aspect can be a motivation for further research on thermal front detection at diurnal time-scales.

Acknowledgements The authors thank Ms. S. K. Anusri (Student, Birla Institute of Technology and Science, Pilani) for helping to develop a standalone CC92 code in python. The authors would like to express their gratitude to Dr. Rashmi Sharma, Dr. B. Kartikeyan, Dr. Arundhati Misra and Dr. Smitha Ratheesh (scientists, Space Applications Centre-ISRO) for useful discussions. We are also thankful to INCOIS for providing the Indian Ocean buoy data sets and IMD for the cyclone track data to validate our results. The authors would like to acknowledge MOSDAC at Space Applications Centre, ISRO for

providing INSAT-3D SST data, MODIS Science team for the MODIS SST data and JPL for MUR SST data.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- Agarwal, N., et al. (2019). Geostationary satellite-based observations for ocean applications. *Current Science*, 117(3), 506.

- Belkin, I., & Cornillon, P. (2003). SST fronts of the Pacific coastal and marginal seas. *Pacific Oceanography*, 1(2), 90–113.
- Belkin, I., & Cornillon, P. (2004). Surface thermal fronts of the Okhotsk sea. *Pacific Oceanography*, 2(1–2), 6–19.
- Belkin, I., & Cornillon, P. (2005). Bering sea thermal fronts from pathfinder data: Seasonal and interannual variability. *Pacific Oceanography*, 3(1), 6–20.
- Cayula, J.-F. (1988). Edge detection for SST images, MS Thesis Dissertation, Department of Electrical Engineering, University of Rhode Island, 91.
- Cayula, J.-F., & Cornillon, P. (1992). Edge detection algorithm for SST images. *Journal of Atmospheric and Oceanic Technology*, 9(1), 67–80.
- Cayula, J.-F., & Cornillon, P. (1995). Multi-image edge detection for SST images. *Journal of Atmospheric and Oceanic Technology*, 12(4), 821–829.
- Cornillon, P., & Watts, R. (1987). Satellite thermal infrared and inverted echo sounder determinations of the Gulf Stream northern edge. *Journal of Atmospheric and Oceanic Technology*, 4(4), 712–723.
- Evan, A. T., & Camargo, S. J. (2011). A climatology of Arabian sea cyclonic storms. *Journal of Climate*, 24(1), 140–158.
- Gangwar, R. K., & Thapliyal, P. K. (2020). Variational based estimation of sea surface temperature from split-window observations of INSAT-3d/3dr imager. *Remote Sensing*, 12(19), 3142.
- Holyer, R. J., & Peckinpaugh, S. H. (1989). Edge detection applied to satellite imagery of the oceans. *IEEE Transactions on Geoscience and Remote Sensing*, 27(1), 46–56.
- Jishad, M., Sarangi, R. K., Ratheesh, S., Ali, S. M., & Sharma, R. (2019). Tracking fishing ground parameters in cloudy region using ocean colour and satellite-derived surface flow estimates: A study in the bay of Bengal. *Journal of Operational Oceanography*. <https://doi.org/10.1080/1755876X.2019.1658566>
- Johns, B., Rao, A., & Rao, G. (1992). On the occurrence of upwelling along the east coast of India. *Estuarine, Coastal and Shelf Science*, 35(1), 75–90.
- Johns, B., Rao, A., Dube, S., & Sinha, P. (1993). The effect of freshwater discharge from the Godavari river on the occurrence of local upwelling off the east coast of India. *Estuarine, Coastal and Shelf Science*, 37(3), 299–312.
- Kilpatrick, K. A., et al. (2015). A decade of sea surface temperature from MODIS. *Remote Sensing of Environment*, 165, 27–41.
- Kostianoy, A. G., Ginzburg, A. I., Frankignoulle, M., & Delille, B. (2004). Fronts in the Southern Indian ocean as inferred from satellite sea surface temperature data. *Journal of Marine Systems*, 45(1–2), 55–73.
- Llewellyn-Jones, D., Minnett, P. J., Saunders, R., & Zavody, A. (1984). Satellite multichannel infrared measurements of sea surface temperature of the NE Atlantic ocean using AVHRR/2. *Quarterly Journal of the Royal Meteorological Society*, 110(465), 613–631.
- Mahadevan, A. (2016). The impact of submesoscale physics on primary productivity of plankton. *Annual Review of Marine Science*, 8(1), 161–184.
- Marra, J., Houghton, R., & Garside, C. (1990). Phytoplankton growth at the shelf-break front in the middle Atlantic bight. *Journal of Marine Research*, 48(4), 851–868.
- Mathur, M., David, M. J., Sharma, R., & Agarwal, N. (2019). Thermal fronts and attracting Lagrangian coherent structures in the north bay of Bengal during December 2015–March 2016. *Deep Sea Research Part II: Topical Studies in Oceanography*, 168, 104636.
- Mathur, M., Haller, G., Peacock, T., Ruppert-Felsot, J. E., & Swinney, H. L. (2007). Uncovering the Lagrangian skeleton of turbulence. *Physical Review Letters*, 98(14), 144502.
- McWilliams, J. C. (2016). Submesoscale currents in the ocean. *Proceedings of the Royal Society a: Mathematical, Physical and Engineering Sciences*, 472(2189), 20160117.
- McWilliams, J. C. (2017). Submesoscale surface fronts and filaments: Secondary circulation, buoyancy flux, and frontogenesis. *Journal of Fluid Mechanics*, 823, 391.
- Price, J. F. (1981). Upper ocean response to a hurricane. *Journal of Physical Oceanography*, 11(2), 153–175.
- Pudov, V. (1992). The ocean response to the cyclone's influence and its possible role in their track formation. In *ICSU/WMO international symposium on tropical cyclone disasters*, 367–376.
- Reynolds, R. W., et al. (2007). Daily high-resolution blended analyses for sea surface temperature. *Journal of Climate*, 20, 5473–5496.
- Sadhuram, Y. (2004). Record decrease of sea surface temperature following the passage of a super cyclone over the bay of Bengal. *Current Science*, 86, 383–384.
- Sengupta, D., Bharath Raj, G., Ravichandran, M., Sree Lekha, J., & Papa, F. (2016). Near-surface salinity and stratification in the north bay of Bengal from moored observations. *Geophysical Research Letters*, 43(9), 4448–4456.
- Suhadha, A. G., & Asriningrum, W. (2020). Potential fishing zones estimation based on approach of area matching between thermal front and mesotrophic area. *Jurnal Ilmu Dan Teknologi Kelautan Tropis*, 12(2), 567–583.
- Thomas, L. N., Tandon, A., Mahadevan, A. (2013). *Submesoscale processes and dynamics* (pp. 17–38). American Geophysical Union.
- Venkatesan, R., Shamji, V. R., Latha, G., Mathew, S., Rao, R. R., Muthiah, A., & Atmanand, M. A. (2013). In situ ocean subsurface time-series measurements from omni buoy network in the bay of Bengal. *Current Science*, 104, 1166–1177.
- Venkatesan, R., Lix, J., Phanindra Reddy, A., Arul Muthiah, M., & Atmanand, M. (2016). Two decades of operating the Indian moored buoy network: Significance and impact. *Journal of Operational Oceanography*, 9(1), 45–54.
- Walton, C., Pichel, W., Sapper, J., & May, D. (1998). The development and operational application of nonlinear algorithms for the measurement of sea surface temperatures with the NOAA polar-orbiting environmental satellites. *Journal of Geophysical Research: Oceans*, 103(C12), 27999–28012.
- Weller, R., et al. (2002). Moored observations of upper-ocean response to the monsoons in the Arabian sea during 1994–1995. *Deep Sea Research Part II: Topical Studies in Oceanography*, 49(12), 2195–2230.
- Zainuddin, M., Saitoh, S. I., & Saitoh, K. (2004). Detection of potential fishing ground for albacore tuna using synoptic measurements of ocean color and thermal remote sensing in the northwestern North Pacific. *Geophysical Research Letters*. <https://doi.org/10.1029/2004GL021000>
- Zavody, A., Mutlow, C., & Llewellyn-Jones, D. (1995). A radiative transfer model for sea surface temperature retrieval for the along-track scanning radiometer. *Journal of Geophysical Research: Oceans*, 100(C1), 937–952.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.